



Chhattisgarh CM Vishnu Deo Sai flags off the country's first hydrogen-powered truck for mining logistics in Raipur on May 10 in the presence of Sanjiv Kumar Katiyar, the Chief Managing Director of Chhattisgarh State Power Generation Company Limited; and Dr Vinay Prakash, Chief Executive Officer of Adani Natural Resources

truck will be useful in mining other metals and minerals. Once hydrogen becomes widely available along major routes, this new -generation vehicle will be capable of longer journeys, significantly reducing carbon emissions across the length and breadth of the country.

Chhattisgarh shows the way

In a global first, Adani Enterprises Limited (AEL) is deploying a hydrogen fuel cell-powered electric tip trailer for mining and logistics operations.

Beyond big cities and bright lights, the initiative is unfolding in Raigarh , Chhattisgarh, a heartland of India's coal mining economy. In financial year (FY) 2024, Chhattisgarh alone contributed over 205 million tonnes (MT), or 20%, of India's nearly one billion tonnes of coal production.

Raigarh will herald a new era of green mining that could serve as a model for other states and countries.

What's the hydrogen truck play like?

In 2022, AEL partnered with Ashok Leyland and Canada's Ballard Power Systems to create this green heavy-duty vehicle. Ashok Leyland is tasked with manufacturing the truck and Ballard is responsible for providing hydrogen fuel cells. Adani New Industries Limited (ANIL), a wholly-owned subsidiary of AEL, is developing the infrastructure for hydrogen fuel.

The truck will be deployed at GP-III coal block in Raigarh for transportation of coal.

What will the truck be like?

This heavy-duty tip trailer will weigh about 55 tonnes and will be equipped with three onboard hydrogen tanks, which have a cumulative capacity to operate around 200 kilometres (km). The contraption features a tipping mechanism, allowing loading and unloading of coal.

The truck will use hydrogen produced through electrolysis. In the fuel cell, hydrogen will react with oxygen to generate electricity, which powers the electric motor. And in a welcome relief, water will be its only emission.

Why does Chhattisgarh matter in India's hydrogen pivot?

Chhattisgarh, one of India's leading mineral-rich states, is a strategic starting point for this initiative. The young central Indian state's economy and employment are inextricably linked to mining. It also faces the environmental consequences of fossil fuel-based industrialisation.

India wants to make a definite statement about its net-zero commitment to the wider world by launching the initiative in Chhattisgarh. The nation seeks to ensure that clean tech doesn't remain confined to cities, but is actively deployed in industrial heartlands, where its impact can be most transformative. If hydrogen-powered vehicles can succeed in Chhattisgarh's rugged mining terrain, they can be scaled across India's coal belts—from Jharkhand to Odisha to Madhya Pradesh.

Environmental and economic benefits

Hydrogen trucks are zero-emission vehicles—their only byproduct is water vapour. They reduce air and noise pollution, making them ideal for use even in environmentally sensitive and urbanised zones. Because hydrogen can be produced from renewable sources, it is a viable sustainable energy alternative.

India leads from the front

In countries such as the US, Germany, Japan, and South Korea, though hydrogen fuel cell trucks are being used, the deployment is of limited scale, mostly in urban logistics and delivery. Companies like Tata and Hyundai Toyota and others have deployed or are developing hydrogen truck fleets.

However, these deployments are primarily for road transportation, not for high-load, off-road industrial use.

AEL will break new ground when it becomes the first company globally to deploy hydrogen fuel cell-powered tip trailers specifically for mining applications.

Once successfully deployed, India can become a global leader in hydrogen logistics and mining innovation.

Significantly, this pioneering initiative will instil confidence in other industry leaders, and a ramp up in scale and operations will reduce transportation costs with hydrogen-powered trucks.